

Skin Lesion Classification using Deep EnsembleNet

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Abstract—Although biopsies have been the standard evaluating metric, specialist rely on pattern and symptoms to make final decision for evaluating skin cancer. Increasing cases of skin cancer and less specialist can impact a lot on the detection. Recently, Convolutional Neural Network (CNN) have emerged as an efficient tool to assist the physician in decision making. Transfer learning is now possible with a large number of pre-trained CNN models. It is frequently necessary to adjust input to suit the input layers in order to use these models. There is quite mismatch between the skin lesion image size and the input layer of different pre-trained models. Resizing corrects the size mismatch; however, it may unintentionally eliminate important characteristics required for precise categorisation. We provide a new EnsembleNet with base MobileNet for the early detection of skin lesions such as nevus and melanoma. Making patches of pictures for input instead of whole picture for classification is being investigated using this method. Also, the other models like denseNet, EfficientNet, ResNet, VGG and Xception Net are studied for the accuracy and loss comparison.

Keywords— *DeepEnsembleNet, Nevus, Melanoma, Skin Lesion*

I. INTRODUCTION

Everyone is now at danger of developing cancer due to a variety of lifestyle variables, such as smoking, drinking alcohol, having strange dietary habits, engaging in physical activity, exposure to radiation from the sun, exposure to radiation from other sources, virus infections, and more. Among them, a spike in incidence of skin cancer accompanied by cell swelling is quite concerning [1]. Two examples of skin cancer, including melanoma and nevus, are used in this work for testing. The two types of skin cancer are more aggressive and life-threatening. Due to their structural and symptomatic similarities, classifying these two categories remains a difficult endeavour. Early diagnosis is treatable and has the potential to save life of patient. Although it is rare, melanoma can sometimes occur in pigmentation such as red, purple, pink, white, or blue [2,3]. Melanoma is often seen in dark or deeper colours. Because of this cancer's propensity for metastasis, or the ability to spread, it is particularly concerning. Although melanoma can appear on the any part of human body, but it typically seen behind the legs [4]. Melanoma is the uncontrolled growth of melanocytes whereas nevi is the melanocytes cluster. According to the research, there might be a 90% reduction in the death rate if skin cancer is detected early. Early detection and categorisation are therefore essential [3, 4]. Numerous studies have been published on the identification, segmentation, and classification of skin malignancies through the use of CNNs, image processing, machine learning, and computer vision

methods. To categorise skin cancer, Esteva et al. [5] employed a CNN, which has the capacity to carry out general and diverse tasks over a broad variety of fine-grained categories. In order to categorise melanomas, Hitoshi et al. [6] developed a semi-automated approach that employed dermoscopic features such as fibrillar patterns, parallel ridges, and parallel furrows as pattern detectors. While Almansour et al. [7] shown a melanoma classification approach with k-means clustering and SVM, Gupta et al. [8] developed the melanoma classification utilising four types of categorisations.

When working with several input data sets or individual picture patches, bootstrapping or bagging has shown to be an effective ensembling strategy. It is possible to assemble ensembles efficiently without increasing the computing workload step by step, as evidenced by recent work [9]. Examining many models, such as VGG19, ResNet, MobileNet, and AlexNet, we discovered that while medical pictures are accessible in a range of sizes, including 1024×1024 , most of them take input image sizes of about 224×224 . This results in scaling being a standard method in many pre-processing techniques. This ultimately results in the removal of several pixels that may have made a major impact. CNN has a harder time learning the features needed for detection or classification when the huge image is down-scaled since there are fewer pixels containing important elements. The spatial link between patches in an ensemble model based on several patches needs to be maintained. The attention mechanism now appears. The successful use of transformer attention in NLP, where many variants are tailored to identification problem such as object detection and segmentation, has led to adoption with attention modelling in computer vision [10, 11]. Getting motivated from this technique we have used a separate parallel channel to keep the feature information intact and added to improve or boost the feature strength.

In this research, we demonstrate the construction of Deep EnsembleNet for melanoma and nevus skin disorders classification, employing MobileNetV2 chosen as a base model because of its least parameters in comparison to other we considered in this paper. Here, we assessed accuracy with loss of model for different model with pre-existing weights. The paper speaks about below points,

- Proposed EnsembleNet with MobileNet as the base classifier.
- Patched based input to each base classifier with intermediate feature fusion matrix.

II. RELATED WORK

Earlier researches [12,13,14] used randomly cropped or enlarged regions of images with the same labelling as the original image in order to train deep networks with fine-grained features. Because feature qualities, and aesthetics may not fully capture the details, this strategy leaves training samples unclear. This problem has been the subject of recent research, which has increased classification accuracy. Bagging [13] is one of the most popular methods for creating ensembles that account for fluctuations in the data; AdaBoost [15] is another. But very few have explained about the patched and intermediate feature matrix approach. An intermediate feature matrix was generated in the suggested architecture by means of a customised bagging algorithm using MobileNetV2 [16]. Also, Huang et al. [17] showed that ensembling requires less computing power since the model converges at several local minima and maintains the model parameters throughout the optimisation process. According to Hara et al. [18], regularisation strategies like dropout fall within the category of ensembling procedures. Tapan et al. proposed a DMF net which can be used for image denoising [19]. They showed that averaging over a network can improve a model's accuracy. In some researches, swap out is an ensembling approach which combines dropout and stochastic depth algorithms. To build the initial basic models that need to be ensembled, all of these methods need in-depth domain expertise. In order to enhance feature selection, recent research has modified the ensembling to satisfy this demand. The spatial relationship between patches may be harmed as a result of the feature matrix's multi-patch ensembling. Learning exact spatial attention is a straightforward solution to this problem, similar to detection and segmentation [20] with slightly different and noisy due to ill-defined label-relevant areas.

III. METHOD PROPOSED

Here the proposed methodology is given in figure 1, and also various other strategies that has been followed to prepare the deep ensembled model has been described which ensures the essential diagnostic information to get intact by eliminating random resizing hence enhance the classification accuracy.

A. Patched RGB skin image

Since arbitrary scaling might cause us to lose important information, these photos are sliced down vertically followed by horizontally, as it is there in Figure 1, to reduce the size of input image patches. The ensembled model is a combination of various pre-trained MobileNet model[21] where each one takes the patches of images. But the experimentation is done in such a way that the performance of patched image has been studied with an incremental basis, where intitial no patch, then two patches and then four patches of the image is done and get input to that many MobileNet models. The final intermediate feature matrix is produced by combining the feature matrices that are generated by each patch of pictures that are fed into the MobileNetV2.

B. Deep EnsembleNet with Attention

Our method concentrates on aggregating intermediate characteristics from many patched inputs to provide a single forecast, in contrast to standard ensembling strategies, which focuses on statistical accumulation of prediction outputs from

different models. This method aids in eliminating the potential for skewed votes during conventional ensembling. The feature matrix obtained by MobileNetV2 with the top three layers freeze, has been put together in accordance with the recommended method in order to correctly identify the image. Here, the bags include patches of various images rather than the complete data set divided into a number of bags. According to Table 1, model MobileNetV2 is selected as it keeps the least parameters compared to other models. To obtain a matrix with a comparable size, we have maintained all of the models the same as MobileNetV2 while we reconstruct the intermediate feature matrix. The complete EnsembleNet with MobileNetV2 (frozen top 3 layers) has been stored throughout model execution, where N: total patches. In this case, N models create the feature matrix using N sets of image patches.

Table 1: Various model's specification

Models	Size (in MB)	Image Size	Parameters (in millions)
Xception [22]	88	299×299×3	22.9
VGG16 [13]	528	224×224×3	138.4
ResNet101 [23]	171	224×224×3	44.7
InceptionV3 [15]	92	299×299×3	23.9
DenseNet121 [24]	33	224×224×3	8.1
EfficientNetB0 [25]	29	224×224×3	5.3
GoogleNet [26]	96	224×224×3	7
MobileNetV2 [16]	16	224×224×3	3.5

Attention describes how the important features are handled minimising the unimportant ones. In accordance with our suggested methodology, we have created many patches from one image in order to extract more pertinent characteristics. We then group the mid model features to get the feature matrix.

Perceptual attention in image processing can be used to enhance relationship of pixel in among patches. The attention module results feature matrix $F_{i,m}(x)$ is:

$$F_{i,m}(x) = ATTN_{i,m}(x) + I_{i,m}(x) \quad (1)$$

where i represents the positions and $m \in 1, \dots, M$ is the number of the channel. $ATTN_{i,m}(x)$: Attention module and $I_{i,m}(x)$: feature matrix generated intermediary. These two feature matrices are superimposed to get the resultant matrix which has enhanced feature information than the earlier one.

IV. TRAINING STRATEGY AND DATA DESCRIPTION

The data collection includes pictures of skin lesions for nevus and melanoma, with corresponding counts of 374 each. The model has tested with one-, two-, and four-patches of images. Skin image datasets gathered from the ISIC2017 data

collection were used for all experiments [27]. Tasks presented are Segmentation, dermoscopic categorisation, and disease categorisation are asked by ISCIC 2017 challenge. Classes like "melanoma," "keratosis" "nevus"—using 374, 254, and 1372 training images, respectively. All the experiments has been taken place with these images.

The Amazon Web Services platform was used for all experiments, and the NVIDIA Tesla architecture with a GPU, CPU x86664, Intel Xenon v4 processor, clock speed of 2.3 GHz, and 4 vCPUs, memory per vCPU is 15.25 (GiB) were used to develop all CNN models. Using the data set, we conducted the experiment using models with preassigned weights such as VGG-19, VGG-16, DenseNet121, ResNet101, EfficientNetB0, and MobileNetV2. As MobileNetV2 is lightweight with fewer parameters, and has higher accuracy, we selected it as our basic classifier for the upcoming two trials. We employed two and four picture

patches using an ensembleNet with an added perceptual attention in the second and third experiments. Various performance metrics to evaluate the model as given in table 3 are performed and also compared.

V. RESULTS

The AUC-ROC plot for the classes Melanoma and Nevus are shown in Figure 2.a and 2.b. The endorsed model has been examined with various other models with specificity, sensitivity, accuracy, and precision are shown in Table 3. It is shown that the proposed has shown promising results than other methodologies. Compared to random resizing, the presented method outperformed previous techniques when the input picture was split into two or four patches to match the input layer. Even though the data set appears modest, it is really five times larger in the quad patch model (four bags for fours patches one for attention), which is used to evaluate [10,5,8,30,31].

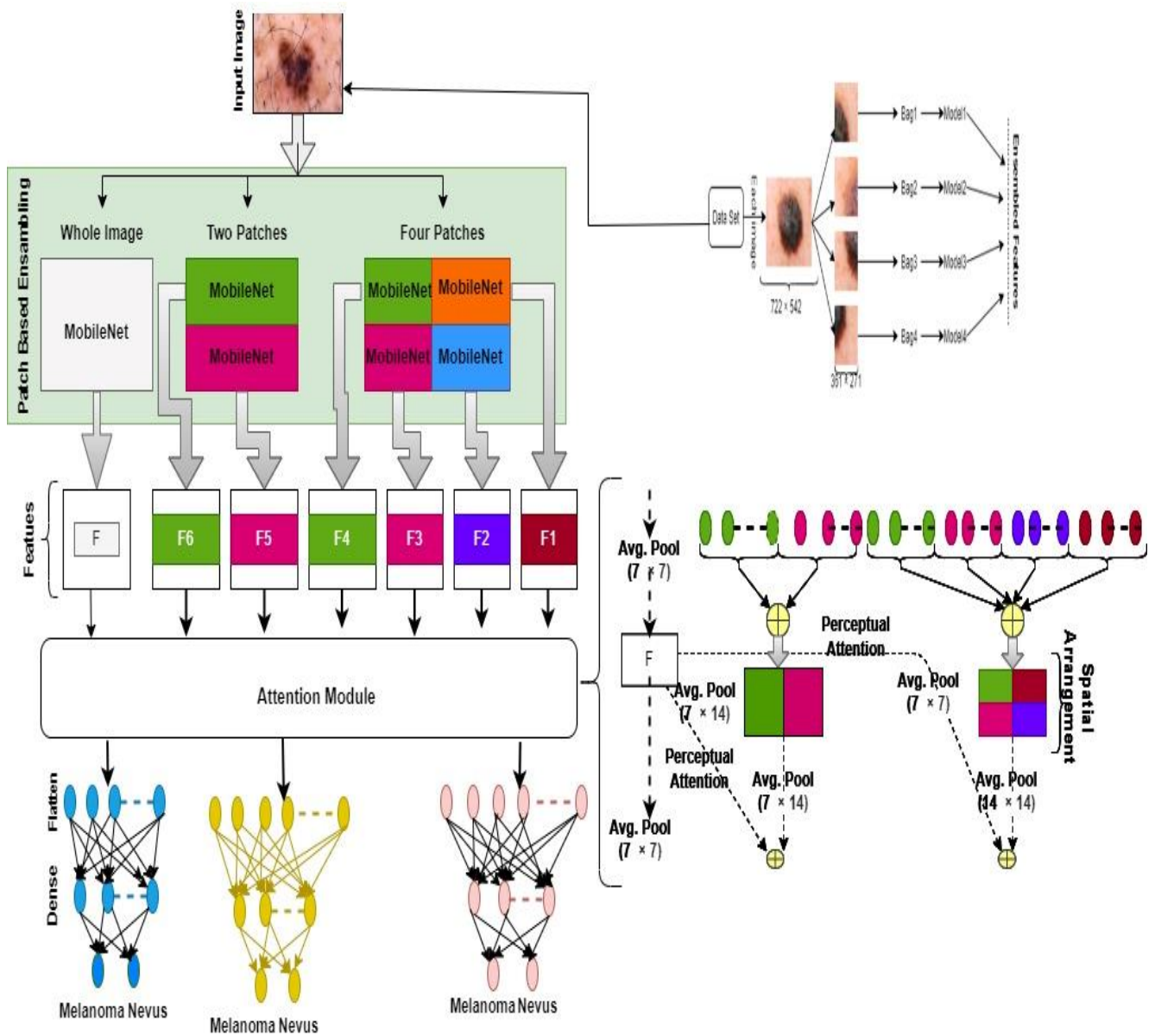


Figure 1: Proposed Methodology for bagging the data set and designing the Deep EnsembleNet Model

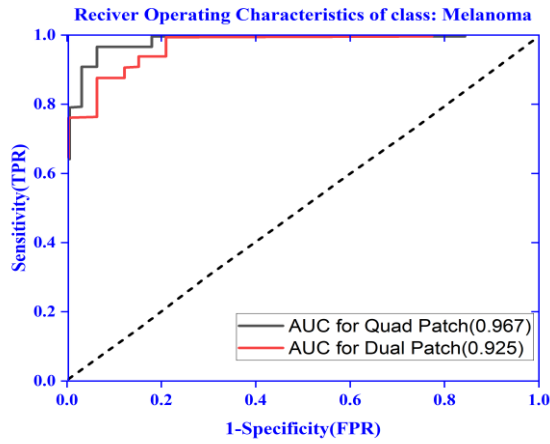


Figure 2.a. AUC plot, class: Melanoma

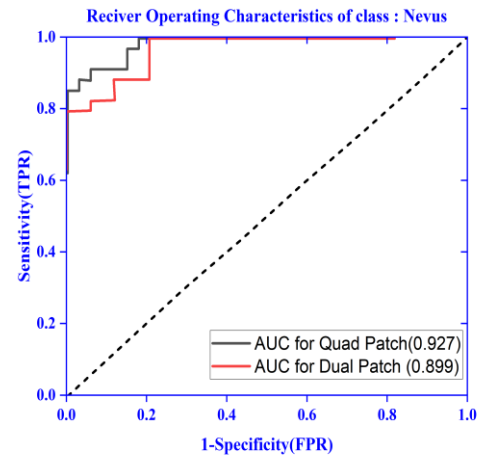


Figure 2.b. AUC plot, skin class: Nevus

Figure 2. plot of AUC

Table 3: Comparison table of proposed methodology with other models

Papers	Methods	Sensitivity	Specificity	Precision	Accuracy
[10]	Texture descriptor	0.62	0.85	0.74	0.76
[10]	Color descriptor	0.74	0.72	0.64	0.73
[5]	Inception V3	0.96	-	-	0.72
[8]	Hybrid texture feature	0.94	0.86	-	0.90
[28]	kNN classifier	0.61	0.80	-	0.67
[29]	ECOC SVM classifier	0.97	0.90	-	0.94
[30]	Illumination correction	0.81	0.80	0.75	0.81
[31]	Optimized NN using PSO	0.86	0.86	-	0.86
[32]	Neural Network	0.72	0.89	0.87	0.805
Proposed Model	Dual Patch	0.967	0.961	0.956	0.920
Proposed Model	Quad Patch	0.981	0.973	0.982	0.953

VI. CONCLUSION

We address two key issues in for dermoscopic image data for skin classification. First, without randomly resizing, we utilise the image's patches (the entire image, two, and four patches) as the input in an incremental way. To preserve the spatial attention, we next apply a perceptual attention module. Which helps in keeping the spatial information intact and improve the classification accuracy of the model. All the three different experiments have been shown in the figure and performance has been evaluated for each one. We show how dividing the input image to patches can lead to higher classification accuracy using DeepEnsembleNet. Here the ensemble technique used is an unique one as earlier ensembling. The two ensembling type know are either boosting or bagging, where boosting based on sequential intermediate output again feed as inpiut to next model until

final output and in bagging we divide the entire data set, but here it comes to learn about the bagging of different features of each image. Hereafter more studies can be conducted out for further division beyond four and it will be learned that at what level we can proceed with the result gets improved.

REFERENCES

- [1] Muhammad Aleem Taufiq, Nazia Hameed, Adeel Anjum, and Fozia Hameed. m-skin doctor: a mobile enabled system for early melanoma skin cancer detection using support vector machine. In *eHealth 360°*, pages 468–475. Springer, 2017.
- [2] Qingli Li, Li Chang, Hongying Liu, Mei Zhou, Yiting Wang, and Fangmin Guo. Skin cells segmentation algorithm based on spectral angle and

- distance score. *Optics & Laser Technology*, 74:79–86, 2015.
- [3] Reda Kasmi and Karim Mokrani. Classification of malignant melanoma and benign skin lesions: implementation of automatic abcd rule. *IET Image Processing*, 10(6):448–455, 2016.
- [4] Robert J Friedman, Darrell S Rigel, and Alfred W Kopf. Early detection of malignant melanoma: the role of physician examination and self-examination of the skin. *CA: a cancer journal for clinicians*, 35(3):130–151, 1985.
- [5] Andre Esteva, Brett Kuprel, Roberto A Novoa, Justin Ko and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *nature*, 542(7639):115–118, 2017.
- [6] Hitoshi Iyatomi, Hiroshi Oka, M Emre Celebi, Koichi Ogawa, Giuseppe Argenziano, H Peter Soyer, Hiroshi Koga, Toshiaki Saida, Kuniaki Ohara, and Masaru Tanaka. Computer-based classification of dermoscopy images of melanocytic lesions on acral volar skin. *Journal of Investigative Dermatology*, 128(8):2049–2054, 2008.
- [7] Ebtihal Almansour and M Arfan Jaffar. Classification of dermoscopic skin cancer images using color and hybrid texture features. *IJCSNS Int J Comput Sci Netw Secur*, 16(4):135–139, 2016.
- [8] Mohd Anas, Kailash Gupta, and Shafeeq Ahmad. Skin cancer classification using k-means clustering. *International Journal of Technical Research and Applications*, 5(1):62–65, 2017.
- [9] Quande Liu, Lequan Yu, Luyang Luo, Qi Dou, and Pheng Ann Heng. Semi-supervised medical image classification with relation-driven self-ensembling model. *IEEE transactions on medical imaging*, 39(11):3429–3440, 2020.
- [10] Jiayuan Gu, Han Hu, Liwei Wang, Yichen Wei, and Jifeng Dai. Learning region features for object detection. In *Proceedings of the European conference on computer vision (ECCV)*, pages 381–395, 2018.
- [11] Han Hu, Jiayuan Gu, Zheng Zhang, Jifeng Dai, and Yichen Wei. Relation networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3588–3597, 2018.
- [12] Xin Lu, Zhe Lin, Hailin Jin, Jianchao Yang, and James Z Wang. Rapid: Rating pictorial aesthetics using deep learning. In *Proceedings of the 22nd ACM international conference on Multimedia*, pages 457–466, 2014.
- [13] Leo Breiman. Bagging predictors. *Machine learning*, 24(2):123–140, 1996.
- [14] Reddy, D. A., Roy, S., Kumar, S., & Tripathi, R. (2023). Enhanced U-Net segmentation with ensemble convolutional neural network for automated skin disease classification. *Knowledge and Information Systems*, 65(10), 4111–4156.
- [15] Yoav Freund and Robert E Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of computer and system sciences*, 55(1):119–139, 1997.
- [16] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4510–4520, 2018.
- [17] Gao Huang, Yixuan Li, Geoff Pleiss, Zhuang Liu, John E Hopcroft, and Kilian Q Weinberger. Snapshot ensembles: Train 1, get m for free. *arXiv preprint arXiv:1704.00109*, 2017.
- [18] Kazuyuki Hara, Daisuke Saitoh, and Hayaru Shouno. Analysis of dropout learning regarded as ensemble learning. In *International Conference on Artificial Neural Networks*, pages 72–79. Springer, 2016.
- [19] Nayak, T. K., Annavarappu, C. S. R., Nayak, S. R., & Gedefaw, B. M. (2023). DMF-Net: a deep multi-level semantic fusion network for high-resolution chest CT and X-ray image de-noising. *BMC Medical Imaging*, 23(1), 150.
- [20] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3431–3440, 2015.
- [21] Nayak, T.K., Rao, A.C.S., Nayak, S.R. (2024). A novel patch-based ensembling approach with perceptual attention for skin lesion classification. *Traitement du Signal*, Vol. 41, No. 5, pp. 2233–2247.
- [22] Ashnil Kumar, Jinman Kim, David Lyndon, Michael Fulham, and Dagan Feng. An ensemble of fine-tuned convolutional neural networks for medical image classification. *IEEE journal of biomedical and health informatics*, 21(1):31–40, 2016.
- [23] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.
- [24] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4700–4708, 2017.
- [25] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning*, pages 6105–6114. PMLR, 2019.
- [26] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed and Andrew Rabinovich. Going deeper with convolutions, *IEEE conference on computer vision and pattern recognition*, pages 1–9, 2015.
- [27] Kiran Ramlakhan and Yi Shang. A mobile automated skin lesion classification system. In *2011 IEEE 23rd International Conference on Tools with Artificial Intelligence*, pages 138–141. IEEE, 2011.
- [28] Noel CF Codella, David Gutman, MEMre Celebi, Brian Helba, Michael A Marchetti, StephenWDusza, Aadi Kallou, Konstantinos

- Liopyris, Nabin Mishra, Harald Kittler, et al. Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic). In 2018 IEEE 15th international symposium on biomedical imaging (ISBI 2018), pages 168–172. IEEE, 2018.
- [29] Ulzii-Orshikh Dorj, Keun-Kwang Lee, Jae-Young Choi, and Malrey Lee. The skin cancer classification using deep convolutional neural network. *Multimedia Tools and Applications*, 77(8):9909–9924, 2018.
- [30] Ebrahim Nasr-Esfahani, Shadrokh Samavi, Nader Karimi, S Mohamad R Soroushmehr, Mohammad H Jafari. Melanoma detection by analysis of clinical images using CNN. In 2016 38th Annual International Conference of the IEEE (EMBC), pages 1373–1376. IEEE, 2016.
- [31] Soumen Mukherjee, Arunabha Adhikari, and Madhusudan Roy. Malignant melanoma detection using multi-layer perceptron with optimized network parameter selection by pso. In *Contemporary advances in innovative and applicable information technology*, pages 101–109. Springer, 2019.
- [32] Serban Radu Stefan Jianu, Loretta Ichim, and Dan Popescu. Automatic diagnosis of skin cancer using neural networks. In 2019 11th International Symposium on Advanced Topics in Electrical Engineering (ATEE), pages 1–4. IEEE, 2019.